NORTHWEST SAMAR STATE UNIVERSITY • COLLEGE OF COMPUTING AND INFORMATION SCIENCES

RESCUE VEHICLE MANAGEMENT SYSTEM (RVMS)

A Two-Platform Fleet Management Solution for the CDRRMO, BFP, CHO, and PNP of Calbayog City

Capstone Final Defense Presentation Pitch **Participating Agencies:** CDRRMO • BFP • CHO • PNP **Academic Year:** 2026–2027

PROBLEM PROBLEM STATEMENT, INTERVIEW FINDINGS, AND RESEARCH LITERATURE

Base from our interviews:

- **Bureau of Fire Protection (BFP):** Relies on physical paper BLOWBAGETS checklists and manual ledger cards, relaying vehicle availability to the LGU via text messages only after maintenance events.
- **City Disaster Risk Reduction and Management Office (CDRRMO):** Lacks standardized daily inspection forms; vehicle readiness is tracked informally through group chats and physical vehicle labeling.
- **City Health Office (CHO):** Depends on informal, driver-triggered verbal defect reporting; dispatchers confirm vehicle readiness by sight since all ambulances share the same compound.
- **Philippine National Police (PNP):** Communicates maintenance concerns strictly through verbal reporting with zero formal fleet documentation at the station level.
- **City GSO Motorpool:** Regularly discovers additional unrecorded mechanical defects during vehicle pre-inspections that agencies failed to report in their initial defect requests.

Consequences from the interview problems:

- **Sudden Mechanical Breakdowns:** Hidden, unrecorded mechanical defects cause sudden vehicle breakdowns while en route to life-or-death emergencies.
- **Fleet Availability Blind Spots:** Dispatchers cannot verify in real time whether an emergency vehicle is genuinely roadworthy, currently dispatched, or stranded in repair, delaying response time.
- **Zero Accountability & Lost Records:** Scattered, unindexed paper files and verbal reporting leave no reliable audit trail of who drove, inspected, or damaged public rescue assets.

Other research studies:

> **1. Mimbala et al. (2024) • Vinay M. G. et al. (2025)**
> Observed that manual logbook-based vehicle monitoring and paper processes in institutional settings lead to inaccurate records, poor data accessibility, and permanent data loss—**with the consequence that agency dispatchers and administrators cannot quickly retrieve reliable vehicle information during critical emergency operations.**

> **2. Crespo del Castillo & Parlikad (2024) • Gallega (2025)**
> Determined that inadequate and disconnected maintenance scheduling severely increases the risk of unexpected vehicle failures and operational downtime—**with the consequence that public emergency vehicles face catastrophic roadside breakdowns when preventative maintenance is not tracked systematically.**

> **3. Carlos Bolaños et al. (2022) • Nerona et al. (2025)**
> Reported that emergency fleet operations without centralized dispatch coordination and automated reporting suffer from poor operational visibility, difficulty consolidating records in local government, and reduced

> accountability—**with the consequence that agency heads cannot verify fleet readiness or hold personnel accountable for vehicle damage and mission delays.**

SOLUTION THE TWO-PLATFORM CENTRALIZED DIGITAL SOLUTION

To resolve documentation gaps and operational risks, the proponents developed the **Rescue Vehicle Management System (RVMS)**, a centralized two-platform digital solution connecting field drivers and agency administrators in real time:

1. Android Mobile Application (For Authorized Drivers):

- **Digital BLOWBAGETS Inspections:** Standardized pre-trip safety checklist (12 standard items + 2 BFP-specific hydraulic/fire pump items) with required remarks for flagged items.
- **Standardized Damage Reporting:** Direct mobile submission of defect descriptions, affected parts, and optional photo attachments.
- **Push Notifications:** Real-time mobile alerts for preventive maintenance schedules and vehicle status changes.

2. Web Administrative Dashboard (For Agency Administrators):

- **Centralized Fleet & Driver Management:** Secure records for vehicle profiles, driver credentials, and automated driver license expiry monitoring.
- **Real-Time Fleet Availability Visibility:** Standardized tracking across four operational states: *Operational*, *Dispatched*, *Not Operational*, and *Under Preventive Maintenance*.
- **Automated Dispatch Logging:** Automatic vehicle status transition to *Dispatched* upon rollout and return to *Operational* with chronological odometer progression.
- **Preventive Maintenance (PMS) & Repair Tracking:** Mileage-based and time-based scheduling with automated Due Soon threshold alerts, paired with GSO and external repair logs.
- **Multi-Agency Data Isolation:** Role-based access separating CHO, CDRRMO, PNP, and BFP records within a shared, secure database.
- **One-Click Reporting:** Instant filtered, printable reports with statistical summaries for audits and agency accountability.

OBJECTIVES PROJECT OBJECTIVES (FROM PROJECT SOURCE OF TRUTH)

The main objective of this study is to develop a Rescue Vehicle Management System. Specifically:

1. **To input vehicle and driver information.** Centralize vehicle technical specifications, assigned drivers, and credentials with automated license expiration tracking.
2. **To track vehicle maintenance.** Standardize digital daily BLOWBAGETS inspections, damage reports, repair records, and scheduled preventive maintenance.
3. **To provide an admin module for dispatch and vehicle management.** Deliver real-time fleet visibility, operational status transitions, and dispatch mission tracking.
4. **To generate relevant reports.** Produce filtered, printable reports with summary statistics for inspections, dispatches, maintenance history, and vehicle status.

DEMO

LIVE SYSTEM DEMONSTRATION • MOBILE APP & WEB ADMINISTRATIVE DASHBOARD